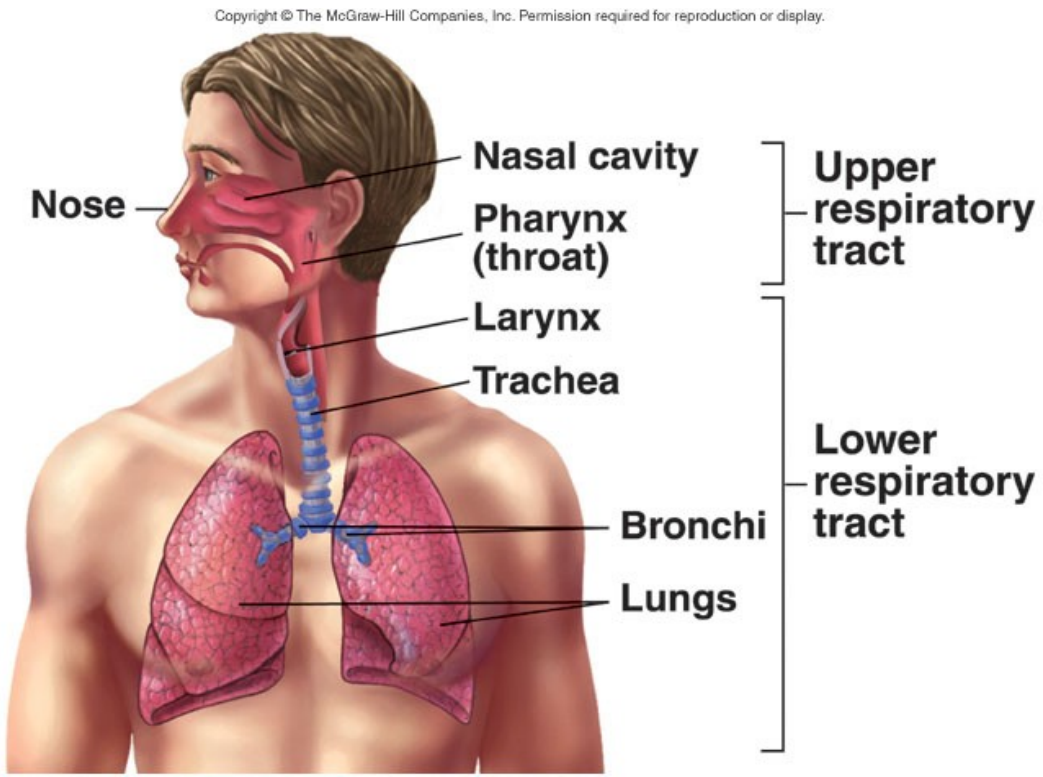
LARYNX, TRACHEA AND BRONCHI

RESPIRATORY MODULE

LARYNX

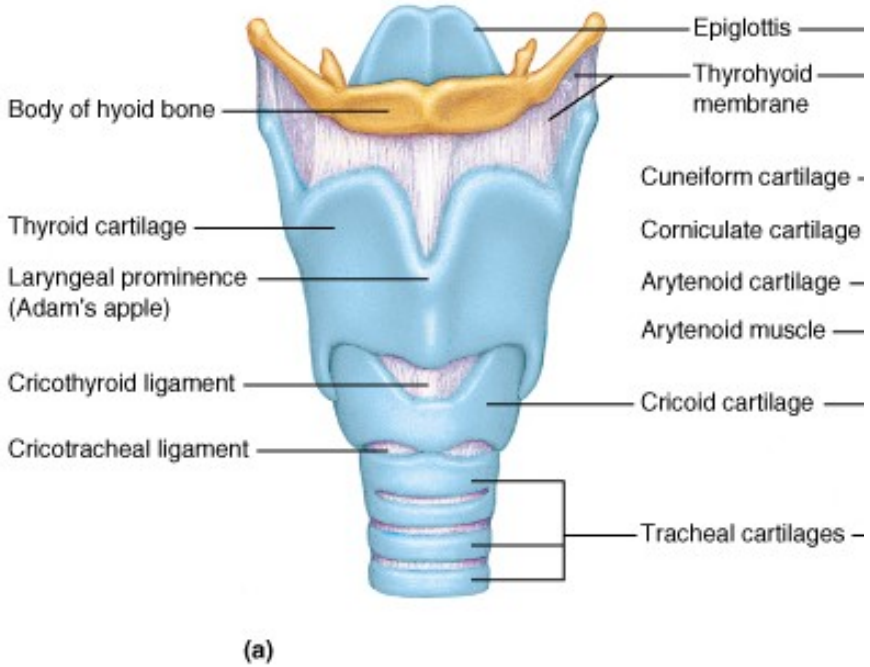
- Larynx is an organ for the production of voice/ phonation, an airway for the passage of air and act as a sphincter at the lower respiratory passages.
- Site: Anterior midline of neck
- Extension: Root of tongue to trachea. From 3rd to 6th C. vertebra.
- Length: 44mm in males and 36 mm in females



CONSTITUTION OF LARYNX

a. Cartilages
b. Ligaments
c. Muscles

- Cavity of larynx is lined by mucous membrane.
- Anterior surface & vocal folds is lined by st. sq. epi
- Remaining larynx is lined by ciliated col. Epi



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LARYNGEAL CARTILAGES

Larynx consists up of six cartilages

<u>Unpaired:</u>

Thyroid

Cricoid

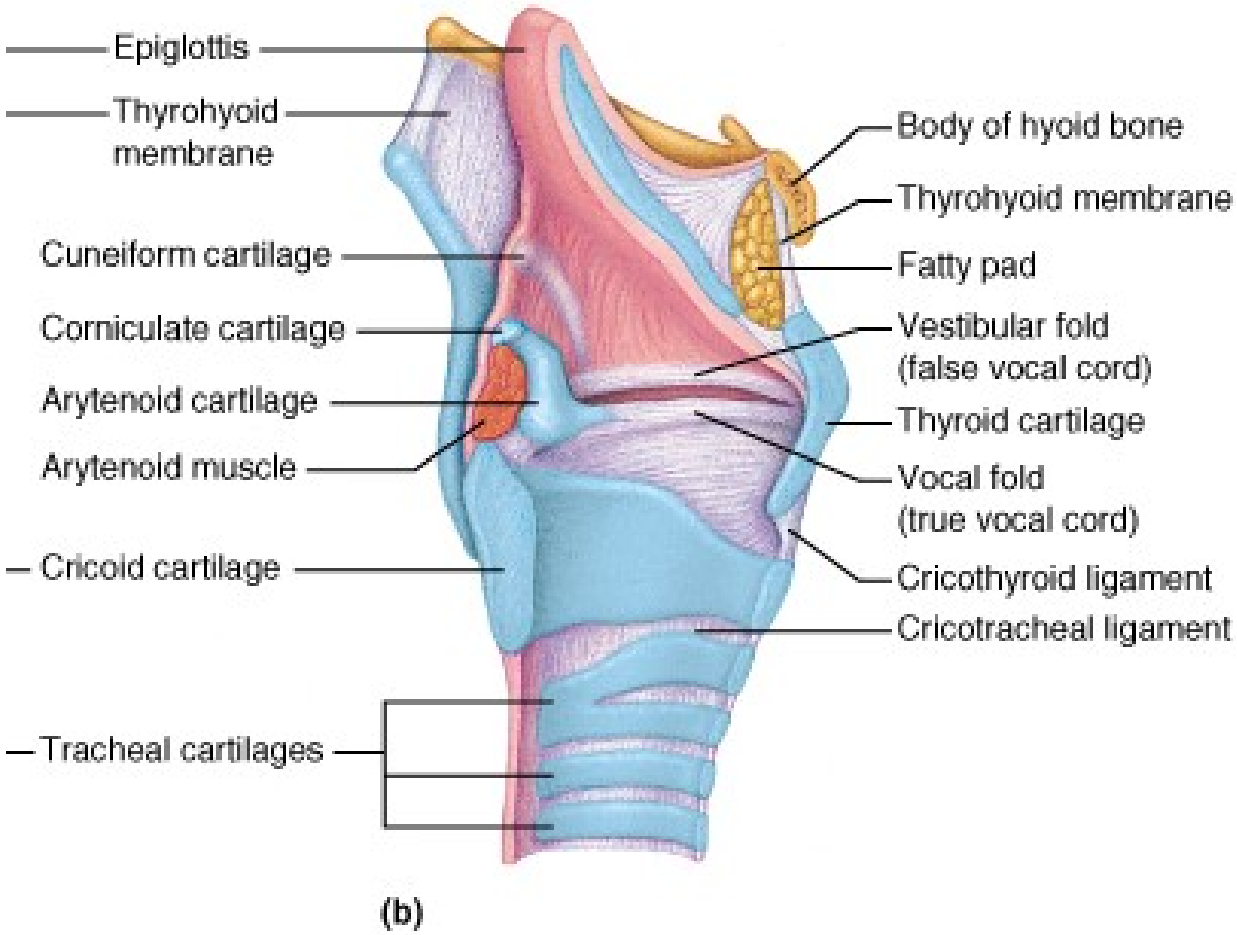
Epiglottic

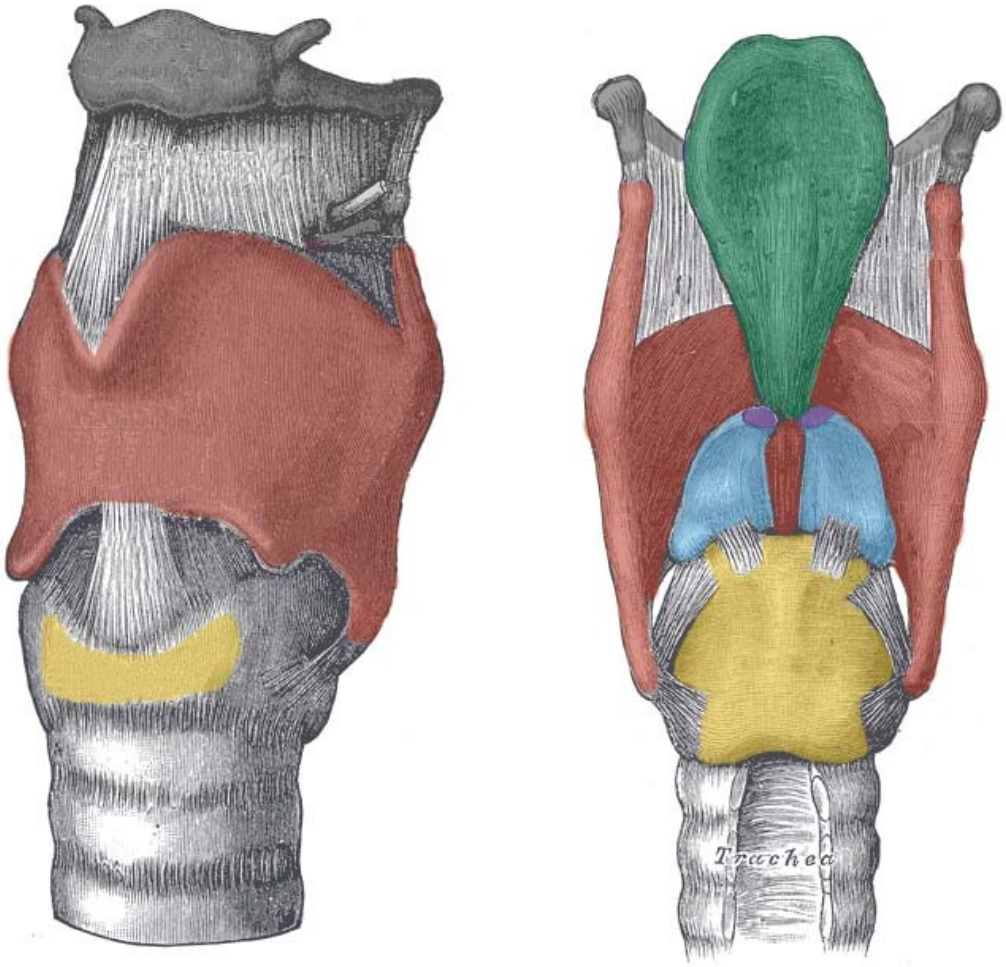
<u>Paired:</u>

Arytenoid

Corniculate

Cuneiform



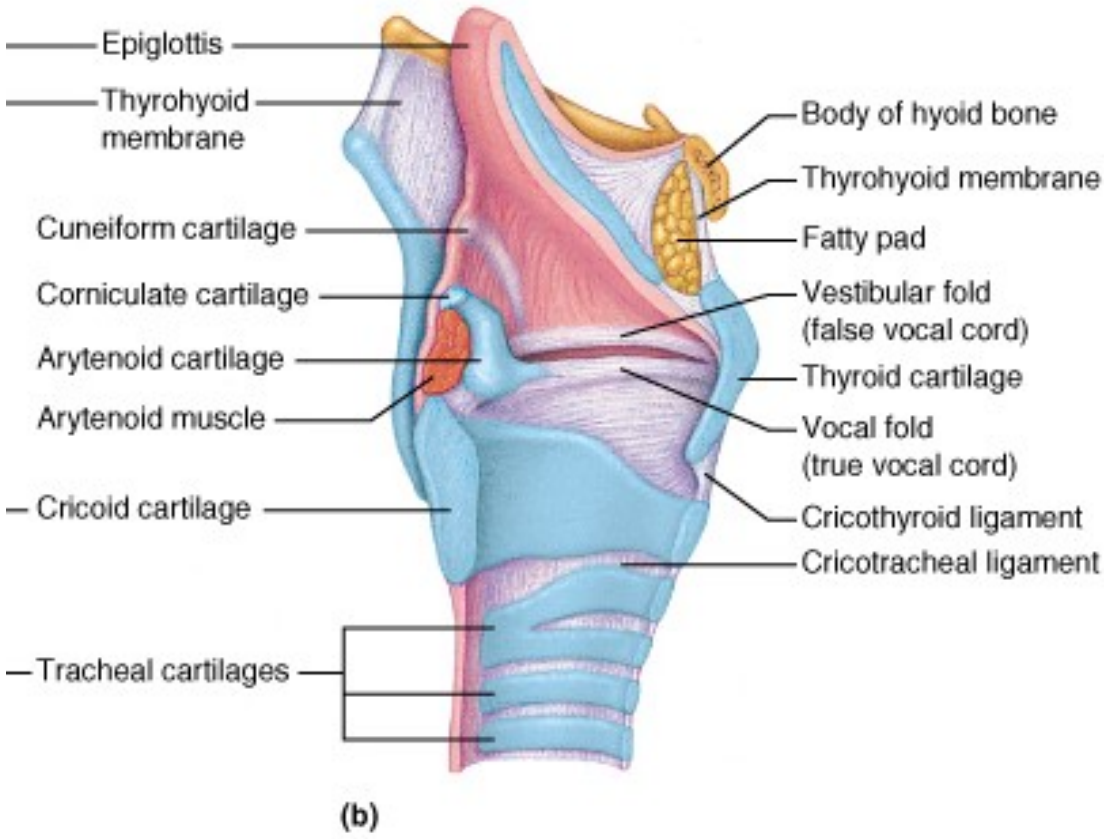


- Epiglottis
- Thyroid cartilage
- Arytenoid cartilages
- Cricoid cartilage
- Corniculate cartilages

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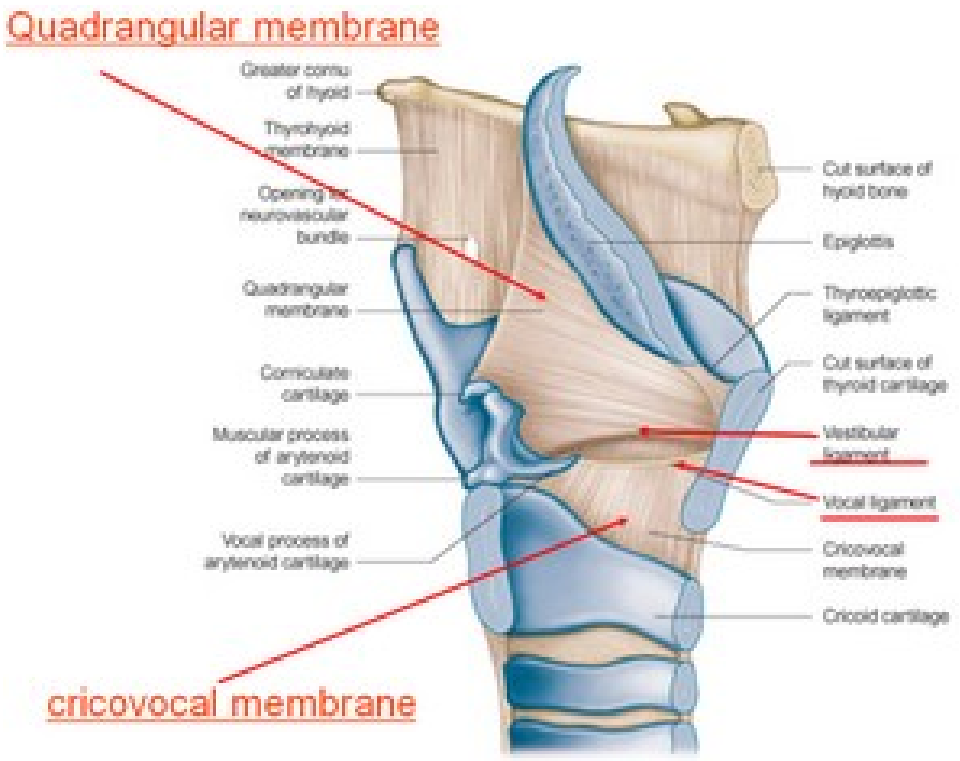
LARYNGEAL LIGAMENTS AND MEMBRANES

i. **Thyrohyoid Membrane;** links thyroid to hyoid;
ii. **Median Thyrohyoid Ligament ;** thickened midline part
iii. **Cricothyroid Membrane;**links thyroid to cricoid;
iv. **Median Cricothyroid Ligament;** thickened midline part
v. **Cricotracheal Ligament;** links Cricoid to first tracheal cartilage



STRUCTURAL LIGAMENTS

a. **Quadrangular Membrane** links Arytenoid to epiglottis
b. **Aryepiglottic Folds** – overlie Quadrangular membrane
c. Lower free edge of Quadrangular membrane is Called **Vestibular Ligament;** deep to Vestibular(False Vocal) Folds
d. **Thyroepiglottic Ligament** links epiglottis to thyroid cartilage.



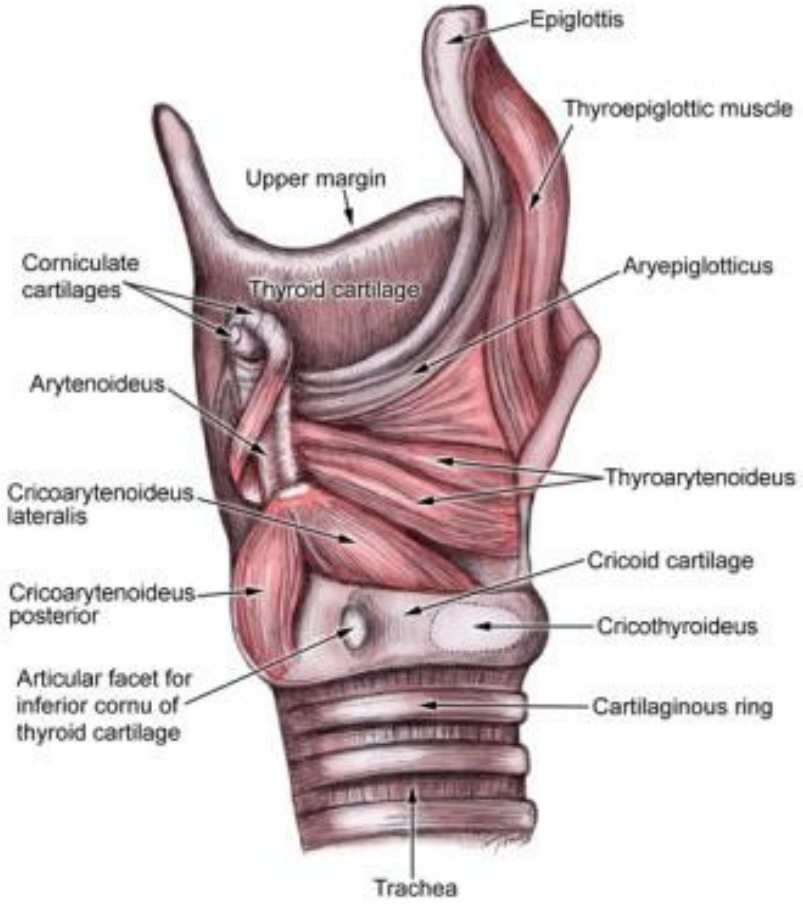
MUSCLES OF LARYNX

EXTRINSIC MUSCLE

i. Suprahyoid (moves larynx during swallowing)

INTRINSIC MUSCLES:

ii. Transverse arytenoid
iii. Oblique arytenoid
iv. Thyroaytenoid
v. Posterior cricoarytenoid
vi. Lateral cricoarytenoid
vii. Vocalis
viii. Aryepiglotticus
ix. Thyroepiglotticus
x. Cricothyroid



ACTIONS OF INTRINSIC MUSCLES

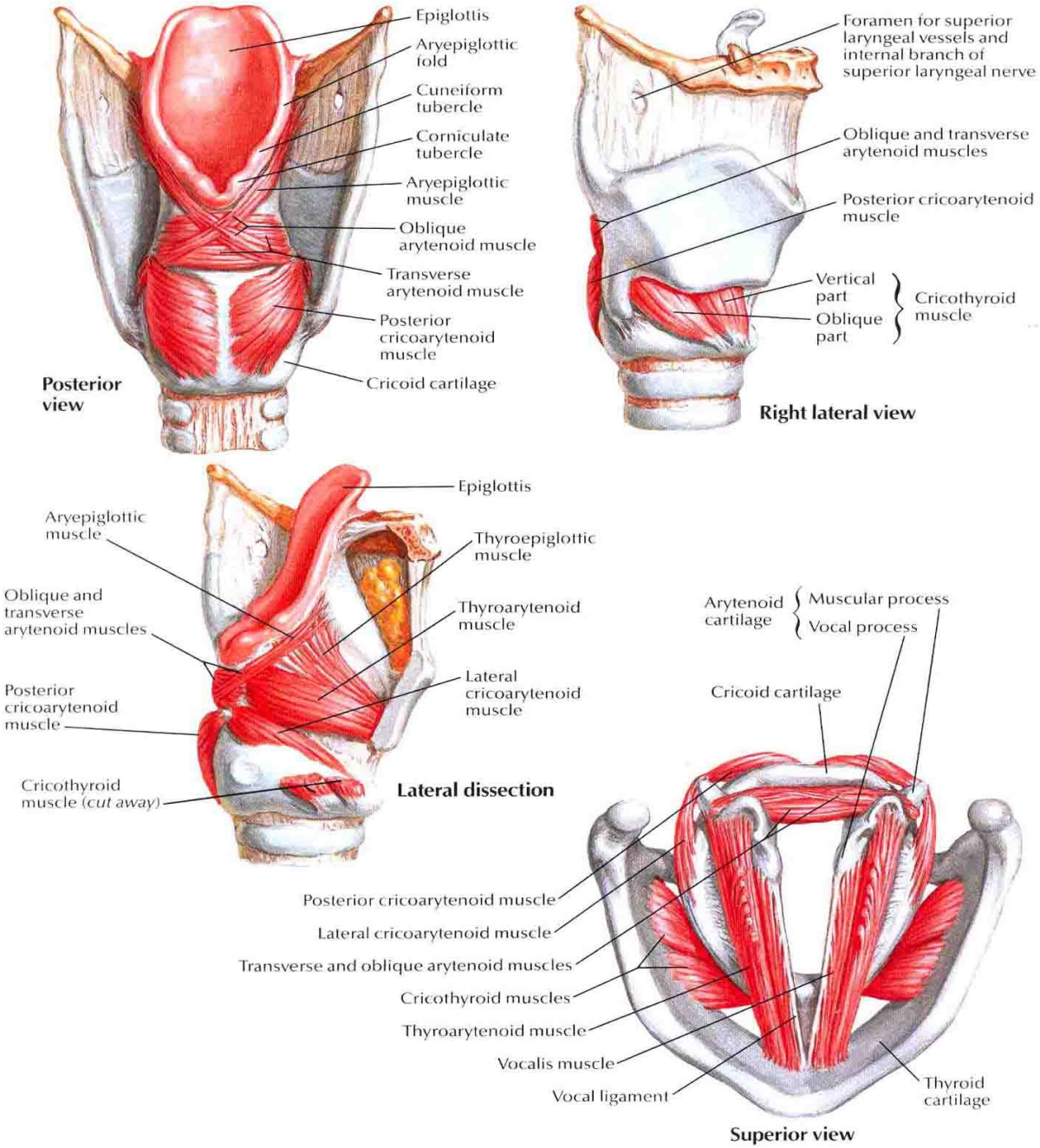
OPENERS OF GLOTTIS	CLOSERS OF GLOTTIS
Posterior cricoarytenoid	Lateral cricoarytenoid
	Transverse arytenoid
	Thyroarytenoid
	Cricothyroid

TENSORS OF VOCAL CORDS	RELAXERS OF VOCAL CORDS
Cricothyroid	Thyroarytenoids
	Vocalis

CLOSERS OF INLET OF LARYNX	OPENERS OF INLET OF LARYNX
Oblique arytenoids	Thyroepiglotticus
Aryepiglotticus	

INNERVATION OF INTRINSIC MUSCLES

- All are supplied by **recurrent laryngeal nerve** except *cricothyroid* which is supplied by *ext. laryngeal nerve*



ARTERIAL SUPPLY AND VENOUS DRAINAGE OF LARYNX

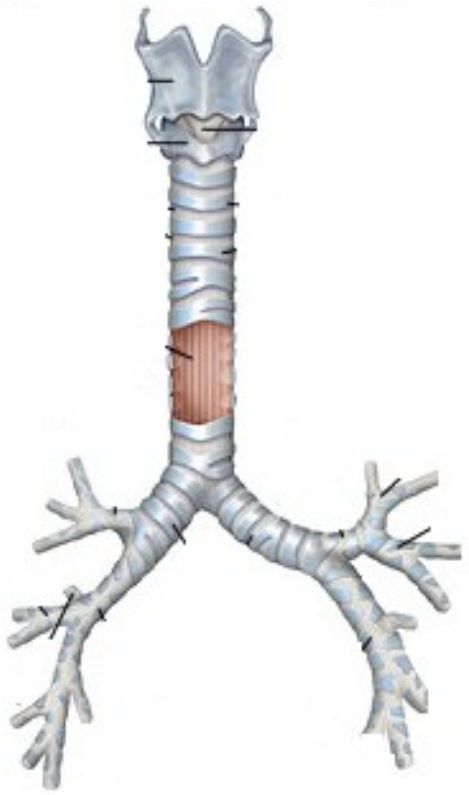
- **Up to the vocal folds**: By superior laryngeal artery Branch Of superior Thyroid artery
- Superior Laryngeal vein drain into superior Thyroid vein
- **Below the vocal folds:** by inferior Laryngeal artery Branch Of inferior Thyroid artery
- Inferior Laryngeal vein drain into inferior Thyroid vein

- **NERVE SUPPLY OF LARYNX:**
- SENSORY SUPPLY; **Internal laryngeal nerve** Supply mucous membrane up to level of vocal folds
- **Recurrent laryngeal nerve** supply below the level of mucous membrane
- **LYMPHATIC DRAINAGE;**
- Above the vocal folds; Anteriosuperior group of deep cervical L.N
- Below the vocal folds; Posteroinferior group of deep cervical L.N

TRACHEA

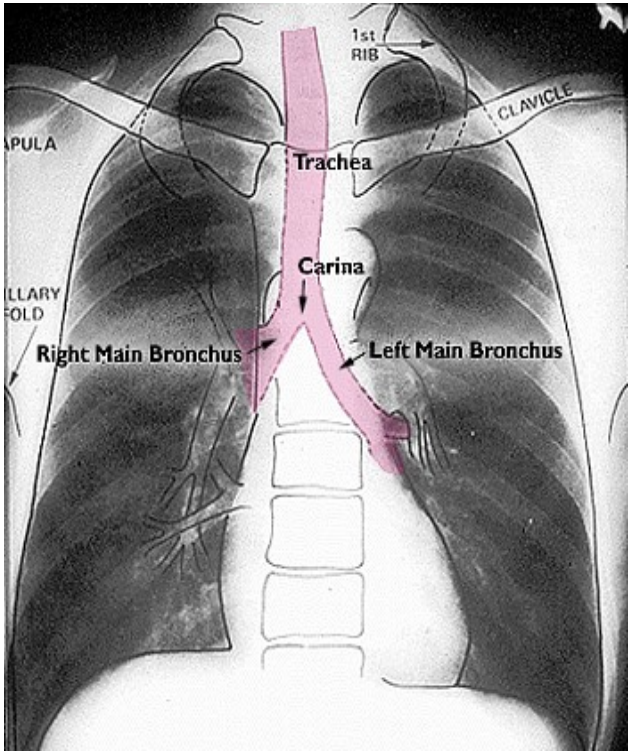
Trachea or windpipe

- is a *cartilaginous* and *membranous* tube, Lies *anterior* to the esophagus
- Extending from the *lower part of the larynx*, on a level with C5-T5
- The trachea is nearly but not quite cylindrical, being flattened posteriorly
- Posterior part of tube lined by **trachealis muscle**
- In the child the trachea is smaller, more deeply placed, and more movable and pliable than in the adult.



- Normally lies at T5,T4 on inspiration
- T6 on expiration
- Lined by **ciliated pseudostratified columnar epithelium.**
- It measures about 9-15 cm. in length
- Outer diameter: 21-27 mm
- Internal diameter: 12-18 mm
- its diameter, from side to side, is from 2 to 2.5 cm., being always greater in the male than in the female.
- It divides into the two bronchi, one for each lung
- Distance from infra cricoid to

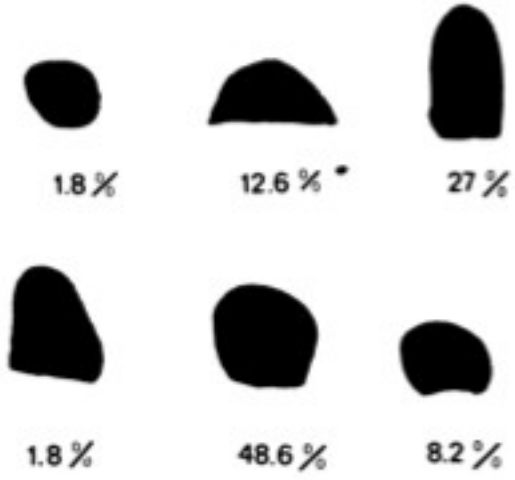
  carina about 11 cm



- Anterior and lateral walls of the trachea supported by 15 to 20 **C-shaped tracheal cartilages.**
- Becomes intrathoracic at 6th cartilaginous ring
- Cartilaginous rings reinforce and provide rigidity to the tracheal wall to ensure that the trachea remains open at all the time

Variable shape

- Usually round, oval, oval with flattened post. border
- Square
- Inverted pear
- Horseshoe



Tracheal Bifurcation

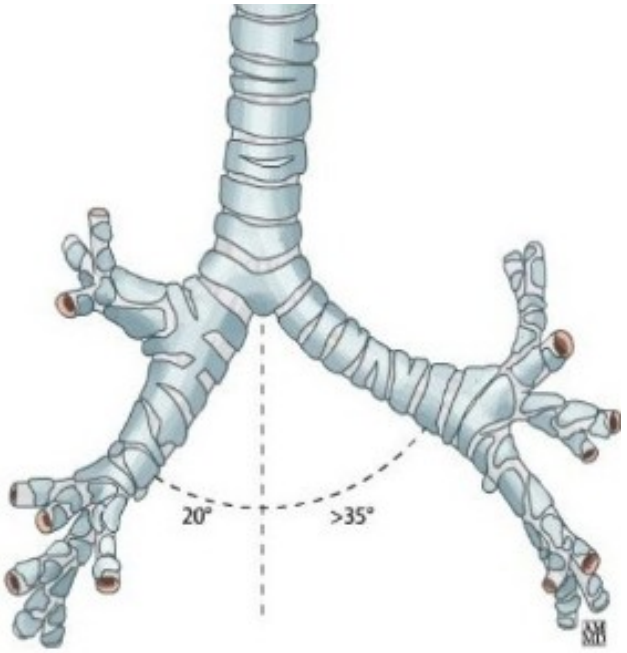
- At the level of the sternal angle, the trachea bifurcates into two smaller tubes, called the **right and left primary bronchi.**
- Each primary bronchus projects laterally toward the each lung.
- The most inferior tracheal cartilage separates the primary bronchi at their origin and forms an internal ridge called the **carina.**

Normal angle: 65°

- 20° to right of midline
- 40° to left of midline

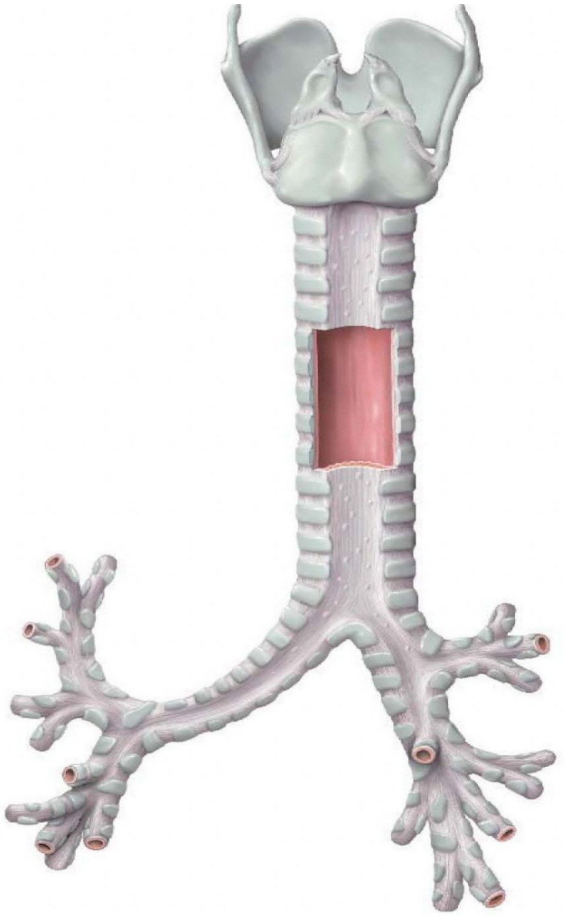
Angle slightly larger + symmetrical in children

Angle increases by 10° - 15° in recumbency



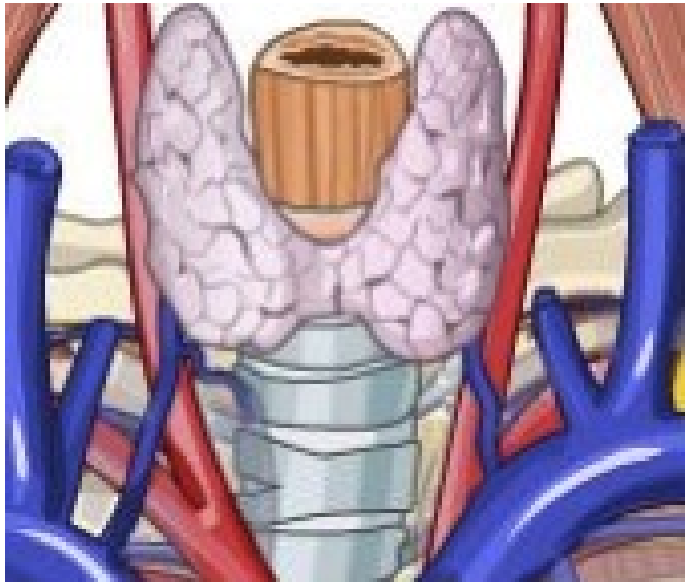
TRACHEALIS MUSCLE

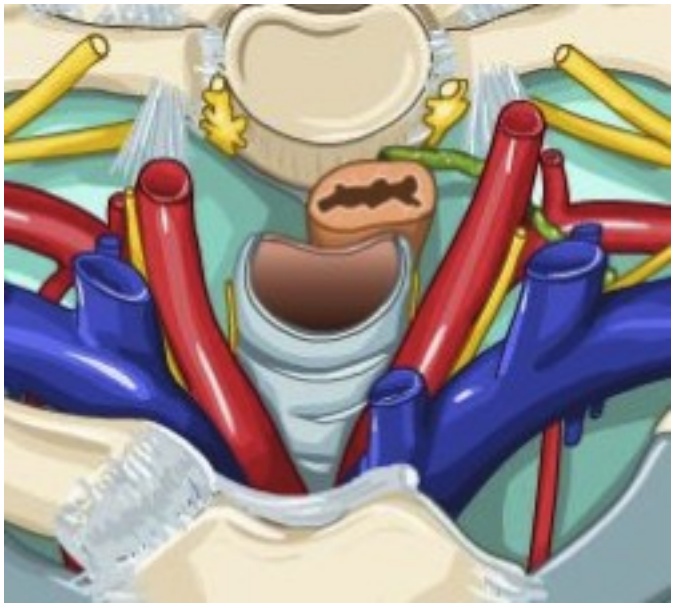
- **Trachealis muscle** is an example of smooth muscle , so it is involuntary muscle. It bridges the gap between free ends of C-shaped cartilages at the posterior surface of the trachea . Its primary function is the constriction and dilation of the trachea while air moves through it. It helps stabilizing the hyaline cartilages.



Relations: Cervical

- **Anterior:**
  - Isthmus anterior to 2nd, 3rd, 4th rings
  - Inferior thyroid veins
  - Strap muscles: Sternohyoid, Sternothyroid
- **Posterior:**
  - Oesophagus, recurrent laryngeal nerves
- **Lateral:**
  - Lobes of thyroid
  - Common carotid artery

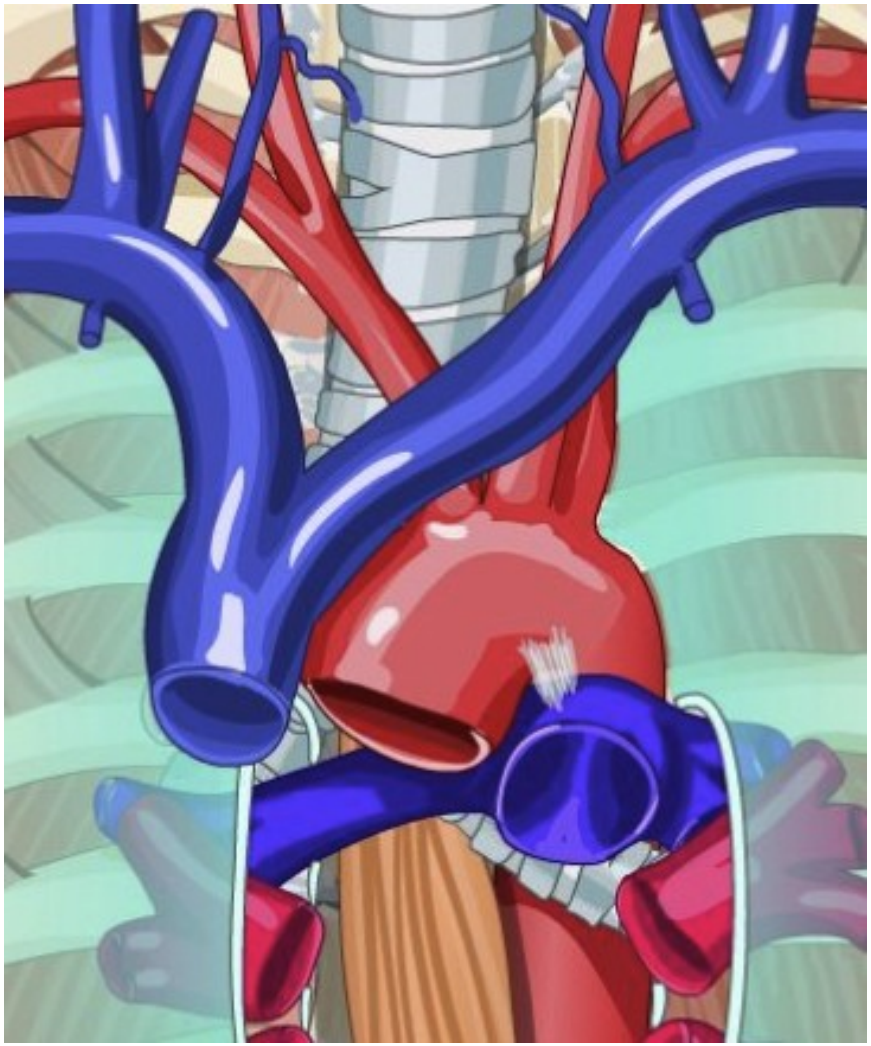




Relations: Thoracic

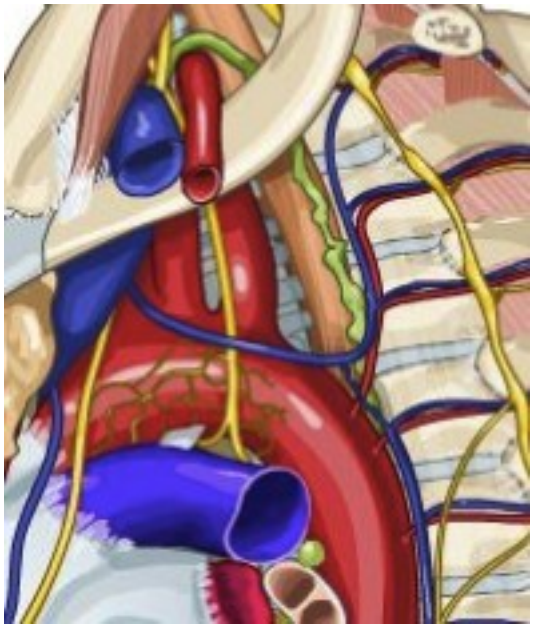
- **Anterior**:
  - Brachiocephalic artery
  - Left common carotid artery
  - Left brachiocephalic vein

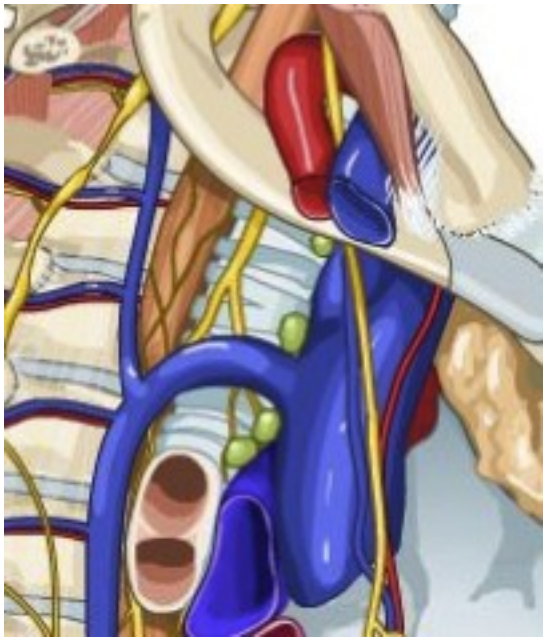
- **Posterior**:
  - Oesophagus
  - Left recurrent laryngeal nerve



Relations: Thoracic

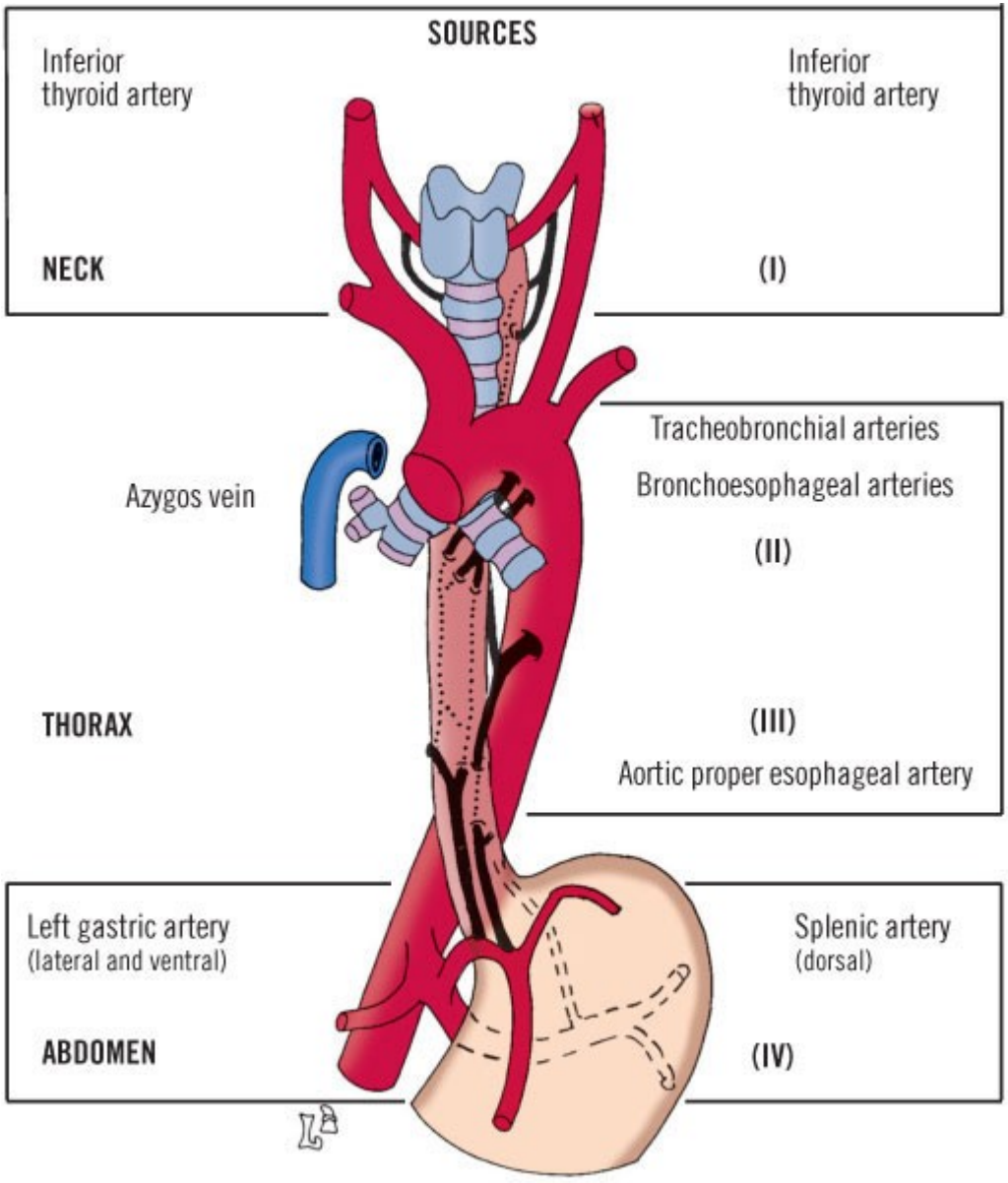
- Left lateral:
  - Arch of the aorta
  - Left common carotid
  - Left subclavian arteries
- Right lateral:
  - Right vagus nerve
  - Arch of the azygos vein
  - Pleura





Blood Supply of the Trachea

- **Upper 2/3** are supplied by the **inferior thyroid arteries**, main source of blood supply
- **Lower 1/3** is supplied by the **bronchial arteries**.
- Ascending and descending branches of these sources anastomose on **surface of trachea**
- Tracheal **veins drain** into **inf. Thyroid venous plexus**



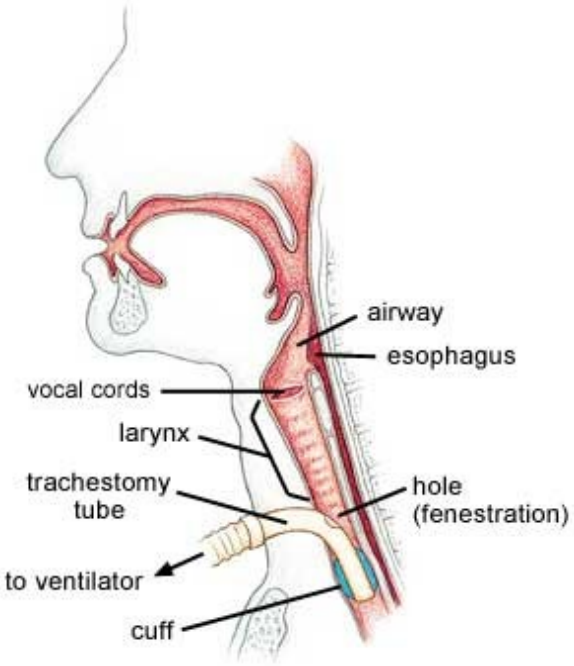
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- **NERVE SUPPLY:** To mucosa , tracheal gland and trachealis muscle is from the **vagus nerves** , the **recurrent laryngeal** nerves and **Sympathetic nerves .**

- **LYMPHATIC DRAINAGE** :Lymph vessels drain into **pretracheal** and **tracheobronchial** lymph nodes which are situated close to trachea from above down ward on side

Tracheostomy / laryngotomy tracheostomy

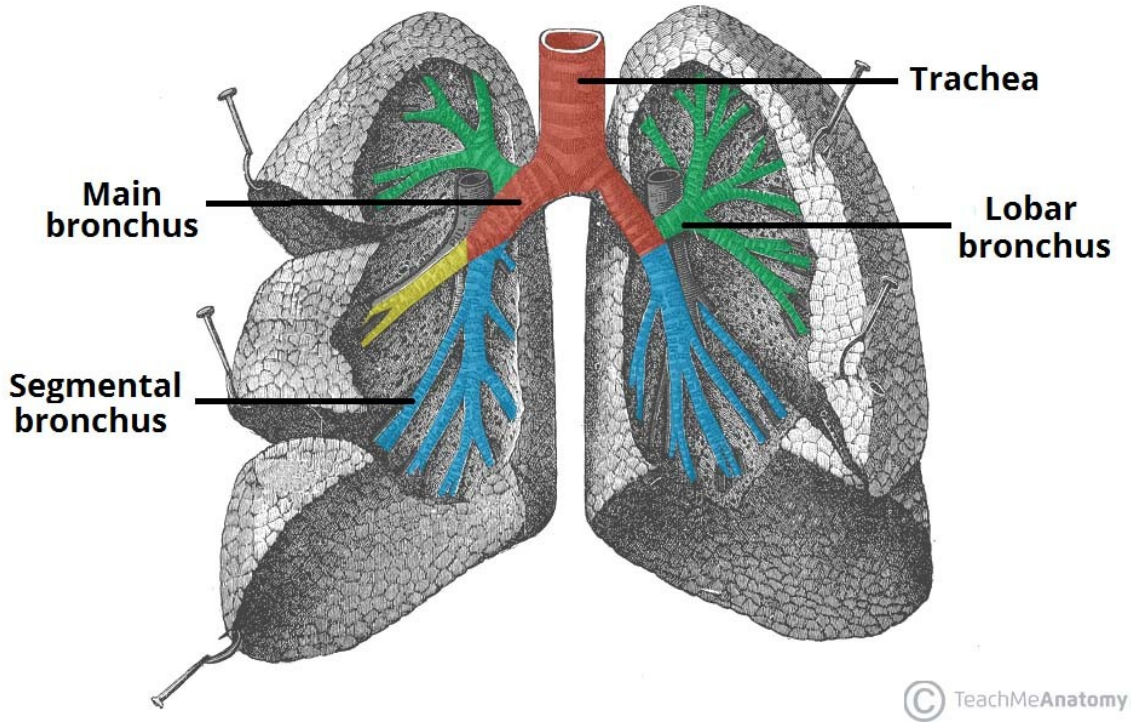
- A surgical procedure ,consists of making an incision on the **anterior** aspect of the neck and opening a direct airway through an incision in **the trachea**. Resulting stoma can serve independently as an airway or as a site for a **tracheostomy tube** to be inserted.
- This tube allows a person to breathe without the use of his / her nose or mouth.
- Both **surgical** and **percutaneous** techniques are widely used in current ENT practice.



Bronchial tree

- Progressively branching of Rt. & Lt. primary bronchi into narrower tubes as they diverge throughout the lungs before terminating in **terminal bronchioles.**
- **Incomplete rings of hyaline cartilage** support the walls of the primary bronchi to ensure that they **remain open** for air entry
- **Right primary bronchus** is **shorter**, **wider**, and more **vertically** situated than the left primary bronchus.
- The **left main bronchus** passes inferolaterally, inferior to the arch of the aorta and anterior to the esophagus and thoracic aorta, to reach the hilum of the lung
- **Foreign particles** are more likely to lodge in the **right primary bronchus**. Specially in children

- Each **main (primary) bronchus** divides into **secondary lobar bronchi**, two on the left and three on the right, each of which supplies a lobe of the lung. Each lobar bronchus divides into several tertiary **segmental bronchi** that supply the bronchopulmonary segments



Blood supply

- The bronchi receive their blood supply from the bronchial arteries, which are branches of the descending aorta. The bronchial veins (which communicate with the pulmonary veins) drain into the azygos and hemiazygos veins.

